PROPOSAL OF AN ELECTRIC TECHNICAL MODEL FOR THE IMPROVEMENT OF EFFICIENCY AND ENVIRONMENTAL SUSTAINABILITY IN SHRIMP PRODUCTION

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Abstract

Ecuador's shrimp sector usually has average efficiency and sustainability due to the pumping system and its energy source. Shrimpers have large expands of land requiring high power pumps, which leads to high fossil fuel (diesel) consumption. CO2 emission rate of 3696 tons per year is obtained, attributed to the pumping system oversized, representing 7.82% of total CO2 emitted by shrimp factories. This fact allows proposing a technical model improving shrimp production efficiency and environmental sustainability by using photovoltaic panels. This research adapts a model that covers Ecuadorian shrimp sector regulations and realities considering 200 and 270 HP pumps and engines respectively, with approximately 14 hours operating time per day, requiring installing 3220 solar panels and 55 batteries. The installation required area for photovoltaic modules is 0.95 Ha. The result obtained priories renewable solar micro generation over fossil fuel, representing a CO2 tons reduction of 7.82% produced by this sector. On the other hand, the proposal economic technical profitability is validated with positive VAN and an internal rate of return approximately 6 times higher than annual interest rate considering the lifespan of the photovoltaic system (20 years), with their respective economic flows by the operation and maintenance of the pumping system.

1 Introduction

Civil society sees as potential benefits of environmental certification in Ecuador, both to protect the environment mainly the mangrove and estuarine ecosystem to increase the competitiveness of Ecuadorian products, the following: Access to markets, reduced costs, social benefits and increased employment. One of the main strengths of the shrimp farms is their water pumping system, which consumes a large amount of diesel for its operation and lubricants for its maintenance, at the same time; it generates CO2 emissions to the environment. This research aims to propose a model of electrical technification to improve the efficiency and sustainability of a shrimp company in Ecuador through the use of photovoltaic panels for shrimp production, which will determine whether or not it is sustainable for this sector.

It begins with a review of the current pumping system of the shrimp farm, the priority equipment for the activities of the production process is evaluated and the equipment of greater criticality is identified, on the parameters established for the analysis. Finally, the model is presented and the economic benefit is presented considering the new Regulation of ARCONEL 050/18.

1.1 Literature review

1.1.1 Aquaculture Farm

Aquaculture activity is defined as that which aims at the reproduction, breeding, cultivation, processing, commercialization and research of hydro biological resources, which can be carried out in continental, inland or coastal waters, defined marine areas and which implies, on the one hand , intervention in the rearing process to improve production and, on the other hand, individual or business ownership of the cultivated stock [1].

Refer to the production of shrimp in captivity, it is a farming activity in aquatic environment, with production and marketing purposes as the final goal, industrialized through technology, however, variables can be found in this regard [2].

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Aquaculture systems in Ecuador can be classified into three methods: extensive, semi-intensive and intensive. Shrimp production in Ecuador is largely semi-intensive, representing less impact on the environment [3].

1.1.2 Water Pumping System

The water requirement is given according to the type of shrimp farming system that is managed: for semi-intensive pools the water exchange is carried out with hydraulic pools, reaching a volume equivalent to half of the peak discharge of the waters of the Guayas River, during the rainy season [4]. Meanwhile, intensive systems require water flows that range between 50 and 60 million litres for each shrimp metric ton. Pumping stations are set of civil works structures, equipment, pipes and accessories that take water directly or indirectly from the supply source and drive it to a storage reservoir or directly to the distribution network [5].

For this method, pumping is done directly from the ocean (more expensive), rivers or aquifers [6]. The procedure to be followed for shrimp production is a circular process, it can be seen that after harvesting the pools are prepared and then filled. A good drying and preparation of the pools contributes to a healthy development of the shrimp, guaranteeing pools free of harmful substances, pathogens and predators that could increase mortality affecting the final yield of the crops [7].

Shrimp farms have high diesel consumption. Within shrimp sector, large part of farms do not have access to electricity grid, for this reason, this fuel is one of the most important energy sources for the operation of machinery such as pumps, generators and even diesel aerators. These devices are used daily and they are essential for the good performance of a shrimp farm. In 2007, the sector had a consumption of 14.1 million gallons of diesel, which meant 12.8 million dollars [8].

The price of a gallon of diesel varies according to the sector for which it will be provided with its respective subsidy and according to the type of diesel used there will be a variation in the price for each of the productive sectors of Ecuador. For the shrimp sector, the sale price of a gallon of diesel for the month of July 2019, according to the company EP PETROECUADOR, at the terminal level for qualified and authorized marketers in the country for the shrimp sector is $ 1.37. On the other hand, the international price per gallon of diesel is $ 3.85 as of July 1, 2019 [9].

The solar energy absorbed by the Earth in a year is equivalent to 20 times the energy stored in all the fossil fuel reserves in the world and ten thousand times higher than the current consumption [10]. Solar radiation can be transformed directly in electricity by means of devices, solar cells or solar batteries that are pure semiconductors doped with minute amounts (1ppm) of other elements, to modify their behaviour.

A typical solar cell is made up of two thin layers of silicon, each with a wire (+ and-) built into it, the end of which sticks out. In one of the layers, some of the silicon atoms are replaced by phosphorous atoms (making it negative or N-type) and in the other, by boron atoms (making it positive or P-type). The junction between two layers creates a potential difference. Solar or visible light induces free electrons to move through the wire of the N-type layer towards the P-type layer, thus producing an electric current that is stored in batteries, to use this energy during the hours without light solar [11].

The autonomous photovoltaic system isolates are used for the most part when the cost of maintenance and installation of power lines is not profitable and it is more feasible to have a photovoltaic solar system [12]. The main applications of these systems are:

- Electrification of homes and buildings, mainly for lighting and low-power electrical appliances.
- Street lighting.
- Agricultural and livestock applications.
- Pumping and water treatment.
- Telephony antennas isolated from the network.
- Signalling and communications.

A pumping system powered by photovoltaic solar energy is similar to conventional systems with the difference that its energy source comes from solar panels; it is a renewable alternative to replace diesel pumps with electric ones connected to its water pumping system [13]. The most important differences between a diesel pump and a solar one are shown in Table 1.

Table 1 Diesel Vs Solar

Pump type	Advantage	Drawbacks
Solar photovoltaic	Unattended operation	High cost of equipment
	No fuel cost	Requires storage for cloudy periods
	Low maintenance	
	Easy installation	Repairs often require trained technicians
	Long service life (20 years)	
	Quick and easy maintenance	Erratic and expensive fuel supply
Diesel and gasoline	Low cost of capital	High maintenance cost
	Widely used	Short shelf life
	They can be portable	Noise and smoke pollution

Source: Practical Action, Technology challenging poverty (2015).

2. Methodology

In the information gathering, the application of surveys was chosen, from which only the information of 7 surveyed shrimp farms can be highlighted, since in each one the visit was made, the water pumping systems are very old and it was not possible to visualize the exact data of nominal power of the equipment for its analysis, in total 30 surveys were carried out with visits to the plant included. The sample of

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work in the project was based on the study [14] that presents an error of 7.6% with a confidence level of 95%. The sample size has a total of 158 shrimp farms distributed in two different strata: provinces and sizes.

2.1 Sample size

The group of shrimp farms are classified as follows:

• Small: from 0 to 49 Ha.

• Medium: from 50 to 249 Ha.

• Large: from 250 to 1,000 Ha.

• Specials: greater than 1,000 Ha.

Of the total shrimp farms, (67.7%) correspond to small farms, 44 (27.8%) are medium farms, 6 (3.9%) represent large farms and 1 (0.6%) correspond to special farms as shown in Table 2.

Table 2 Base sample of surveyed shrimp farms

Province Size	Guayas	Manabí	Santa Elena	El Oro	Esmeraldas	National Total
Little	29	27	1	37	13	107
Medium	28	4	0	10	2	44
Big	4	0	0	1	1	6
Specials	1	0	0	0	0	1
Total	62	31	1	48	16	158

Source: MAGAP, 2015

Elaboration: Sarmiento, J. (2018)

An analysis of diesel consumption was carried out for the shrimp sector, where it was found that the process with the highest demand for this fuel is water pumping (84.2%), compared to electricity generation (7.3%), aeration of swimming pools (8.3%) and use of blowers (0.11%) [14].

2.2 Preparation of the Shrimp Farms survey

The survey includes the following group of questions related to the water pumping system within the farms, posed with the objective of collecting specific information.

The survey reflects three sections detailed below:

• The first section of the questionnaire has questions about the mechanical characteristics of the water pumping system. It focuses on the machinery used in pumping, based on this, to be able to understand the technical requirement of each shrimp farm. These tables take specific information, for example, the technical characteristics, working hours and age of the machinery.

• The second section focuses on the physical characteristics of the shrimp farm that influence the pumping system, such as the height of the suction zone to the reservoir channel and the characteristics of the pipes.

• In the third section, information is collected about the costs related to the maintenance, repair of the machinery, and the transportation of fuel to the shrimp farm. This is important for the cost balance between a renewable technology and the current conventional technology used.

The group of people to whom this survey was used were contacted by telephone from September 1, 2016 until May 2017.

For this design, with renewable energies, the solar panels must supply the energy necessary for the motor to operate the pump and for this, in turn, to enter the water into the pools of the shrimp farm. As first step, electric motor that meets the power requirements of the pump. Subsequently, a high speed soft starter and a static power converter are needed that meet the motor's requirements for voltage and power rating; finally, the accumulation batteries and the nominal power of the solar panels are sized.

The cost of production with solar energy and diesel can be shown in the following table and additionally the amount of CO2 that is emitted into the environment can be compared.

Table 3 Production cost data

Power source	Cost	Emission of CO_2
Diesel	0.18 $/kwh	0.709 kg/kwh
Solar energy	0.03 $/kwh	0 kg/kwh

Elaboration: Sarmiento, J. (2018)

3

Results

Table 4 Results Obtained from the Intersection between the Curves

Shrimp Farm	Pump Brand	Number of Pumps	Pump Diameter (inches)	Pump Height (m)	Flow (m^3/s)	Efficiency (%)
	Delta delfini 36''	5	36	4,75	2,69	75%
A	Delta delfini 28''	2	28	5,2	2,69	80%
	Etec 48''	3	48	4,01	4,80	84%
	Delta delfini 36''	5	36	4,80	2,69	76%
B	Machaleña 36''	4	36	3,90	2,74	80%
	Etec 48''	5	48	4,00	4,80	80%
C	Delta delfini 36''	3	36	3,50	2,80	66%
	Etec 36''	2	36	4,00	2,81	84%
D	Delta delfini 50''	4	50	4,00	5,38	72%
	Machaleña 36''	4	36	3,90	2,75	80%
	Delta delfini 24''	2	24	5,00	1,31	74%
	Delta delfini 36''	1	36	4,75	2,70	75%

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E	Delta delfini 36''	5	36	4,50	2,71	75%
	Delta delfini 36''	1	36	5,00	2,67	79%
	Delta delfini 36''	1	36	4,70	2,70	75%
	Nacional 36''	1	36	4,60	2,69	81%
F	Delta Delfini 36''	2	36	4,70	2,70	75%
	Delta delfini 36''	2	36	4,70	2,70	77%
	Delta Delfini 36''	2	36	4,40	2,70	74%
G	Delta Delfini 36''	1	36	4,30	2,70	76%

Once the system curve was obtained for all the pumps in the shrimp farms, these were intersected with the characteristic curve of each pump. From these calculations, the results were obtained in terms of pumping head (Hb), flow Q and efficiency η, as shown in Table 4.

3.1 Design of the Pumping System with Photovoltaic Solar Panels

To define the design of a photovoltaic pumping system, the Shrimp Farm C in the sample is chosen as an example because it has three pumps of equal power, unlike the pumping stations of the remaining shrimp farms whose pumps have very different powers between them. These farms, like all those that are part of the sample, do not have access to the electrical distribution network at the 13.8kV level. As can be seen in Table 5, the motor-pump system in the Shrimp Farm C is 20 years old, so a change of machinery is required. Therefore, within this study, the substitution of the diesel system for an electric one is proposed in the case in which a medium voltage line can be built to feed the pumping system and obtain the tariff implemented by the government in ARCONEL 050 / 18, for this proposal solar energy will be used for a single pump, which is the same for the remaining pumps in the station. The purpose of the photovoltaic solar pumping system is to generate clean energy for the shrimp farm through which it would stop consuming 103,432 gallons of diesel per year, representing 698 tons of CO2 per year

Table 5 Characteristics of the Diesel Pumping System of the Shrimp Farm C.

Shrimp Farm	Location	Extension (ha)	Size	Pump type	Number of Pumps	Age of the bomb (years)	Calculated Pump Power (hp)	Calculated Motor Power (hp)
C	Isla Quiñonez -Guayas	304	Big	Delta 36 "	3	20	200	269

3.2 Financial Analysis of the Efficient Pumping System Project

This financial analysis is intended to generate quantitative information on economic and financial aspects that allow us to assess the economic feasibility of different project alternatives, based on their profitability. We must determine the cash flow and the cost of capital or discount rate of the project to verify if it should be invested in that of the shrimp farmers.

In this section an evaluation of the alternatives for the Farm C pumping system is presented, due to the age of the current diesel motor-pump system that it has is 20 years, therefore it requires the renewal of the equipment, so that the farm continues with its normal operation and shrimp production. The pump station contains three pumps, where the results obtained from oversizing require a power of 200 hp for each of the pumps and 275 hp for each motor. Since the pumping system of this shrimp farm has three identical pumps in terms of the power required, the replacement of a single pump has been taken into account for this analysis. For this reason, the behaviour of the results obtained for one pump will be the same for the two remaining pumps.

The Proposal of a Model of Electrical Technification for the Improvement of Efficiency and Environmental Sustainability in Shrimp Production; proposes the change of the conventional pumping system of the shrimp farm (diesel) for a photovoltaic system in order to increase the production of the farm, consists of investing again in the purchase of equipment (motor, pump)

Table 6 Photovoltaic Pumping System

Equipment	Quantity	Potency Hp(Kw)	Capacity (Ah)	Voltage (V)
Electric bomb	1	200 (149)	-	-
Electric motor	1	270 (201)	-	380
Speed Soft Starter	1	300 (220)	-	460
Static Frequency Converter	1	201 (150)	50-1500	380
Static Frequency Converter	2	40 (30)	50-1500	380
Accumulation Batteries	55	-	1280	48
Solar modules	3220	0,42(0,310)	8,42	36,8

3.3 Budgets for the Photovoltaic Pumping System

In this alternative, a scenario is presented where the price of the solar pumping system is made up of the loaded value of the equipment plus other additional costs that are mentioned below. It should be specified that this budget was determined based on current tariff conditions in Ecuador such as VAT, which has a value of 12%.

Equipment such as static energy converters and storage batteries must be imported; therefore, import taxes must be increased at their nominal price. Within Resolution 011-2015 of the Ministry of Foreign Trade, a 45% tax is stipulated for batteries, however, it is 0% for rechargeable batteries, as is the case with the batteries that are proposed for this system. For static energy converters a 15% duty must be paid.

Table 7 shows in detail the budget for the photovoltaic pumping system project for Farm C

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Table 7 Photovoltaic Pumping System Costs

Item	Description	Unit	Quantity	Unit cost	Total cost
1	Electric bomb	u	1	$17100,00	$17100,00
2	Electric motor	u	1	$20358,87	$20358,87
3	Soft starter	u	1	$6817,13	$6817,13
4	150kw inverter	u	1	$65931,79	$65931,79
5	30kw inverter	u	2	$32965,90	$65931,80
6	Batteries	u	55	$14202,00	$781110,00
7	Solar panels	u	3220	$444,60	$1431612,00
8	150kw inverter	u	1	$8675,24	$8675,24
9	30kw inverter	u	2	$8672,24	$17344,48
10	Photovoltaic panel structure	u	230	$593,71	$136553,30
11	Battery room	u	1	$2052,00	$2052,00
12	Installation labor	u	1	$22800,00	$22800,00
13	Equipment transport	u	5	$2500,00	$12500,00
14	Electric material	u	1	$22800,00	$22800,00
15	Others	u	1	$500,00	$500,00
16	Operation	gb	1	$634965,46	$634965,46
17	Maintenance technician during the useful life of the equipment	gb	1	$634965,46	$634965,46
18	Maintenance during the useful life of the equipment	gb	1	$138686,65	$138686,65
19	Person in charge of cleaning solar panels during their useful life	gb	1	$94024,85	$94024,85
20	Maintenance of the motor-pump during its useful life	gb	1	$44661,80	$44661,80
21	Repair during its useful life	gb	1	$13603,95	$13603,95
22	Repair of the motor-pump during its useful life	gb	1	$13603,95	$13603,95
Total project cost					$4186598,73

The proposal for an electrical technification model to improve efficiency and environmental sustainability in shrimp production has a payback in the third year of implementation of the photovoltaic project, with an annual interest rate of 10%; which gives us a NPV of $ 12,392,297.19 and an IRR of 57% in a period of 20 years, which is the useful life of all the equipment of the photovoltaic system, including expenses for operation and maintenance of the pumping system

4 Conclusion

The objective of this study was to propose a model of electrical technification to improve the efficiency and sustainability of shrimp farms in Ecuador, for which a case study was made. From 55 equipments that make up the pumping stations of the shrimp farms for the analysed sample, it was found that 35 pumps have a power greater than that demanded to satisfy the requirements of the farm's pools.

It has been determined that for a photovoltaic system to supply a 200 hp pump and 270 hp motor, with an operating time of approximately 14 hours a day (7 hours a day and 7 hours a night), it is necessary to install 3,220 panels of 310 Wp and 55 batteries of 1,280 Ah. The area required for the installation of the photovoltaic modules is 0.95 Ha, which does not represent an inconvenience for the shrimp farms in the sample, since they have a land area greater than 250 Ha.

In order to minimize the cost of the photovoltaic solar system, it is best to take the option of installing a hybrid system, it works during the day through photovoltaic panels and at night with the traditional diesel system. In this way, the price of the system is reduced because the use of batteries would be eliminated and the number of panels would be much less than that established in the referential budget with a value of $ 2,059,731.76 with a diesel subsidy.

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